Detecting Weak Underwater Targets in Hyperspectral Imagery via Physics-Aware Residual Reasoning

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***Abstract*—Detecting weak underwater targets in hyperspectral imagery (HSI) is challenging due to strong background interference, water-induced spectral distortions, and the inherently low responses of submerged materials. Current methods typically rely on target-oriented strategies, attempting to strengthen weak signals by embedding limited target priors. However, they remain constrained by two critical issues: the scarcity of target pixels and the absence of physically grounded constraints, both of which undermine weak-signal preservation and result in unreliable detections. To address these limitations, this study introduces rejection-guided residual reasoning network (R^3Net), a rejection-guided residual reasoning framework built upon the principle that *rejection is the best protection*. The main idea is to shift focus from target-oriented formulations toward a background-oriented paradigm that leverages the abundance of background information as a stable and reliable reference. Building on this idea, R^3Net first establishes a background-oriented representation by explicitly suppressing target contributions, allowing the residuals between the input and its reconstruction to naturally reveal candidate target signals. Since such residuals inevitably encompass background noise and nontarget weak responses, they are subsequently refined through a physics-aware reasoning stage grounded in the underwater imaging model, which validates candidate signals based on physical consistency. In this way, R^3Net unifies residual learning with physics-guided refinement, enabling robust preservation of weak targets while effectively suppressing false alarms, thus achieving reliable and physically interpretable underwater detection. Extensive experiments on real-world datasets demonstrate that R^3Net outperforms state-of-the-art (SOTA) methods in both detection accuracy and robustness. All data and source code used in the experiments will be publicly available at https://github.com/qjh1996/HUTD**

***Index Terms*—Hyperspectral imaging, physics-informed model, residual reasoning, target rejection, underwater target detection.**

I. Introduction

HYPERSPECTRAL remote sensing [1], [2], [3], [4], [5], [6], [7], [8] has recently emerged as both a powerful approach and an active research focus for underwater target detection [9], [10], [11], [12]. Compared with visible and multispectral sensors that suffer from severe absorption and scattering in water, hyperspectral imagery (HSI) provides continuous spectral measurements, enabling more reliable separation of targets from backgrounds. As shown in Fig. 1, conventional optical imagery lacks distinct spatial cues, whereas HSI supplies characteristic spectral signatures that preserve discriminability under such conditions. These advantages have driven the development of hyperspectral underwater target detection (HUTD).

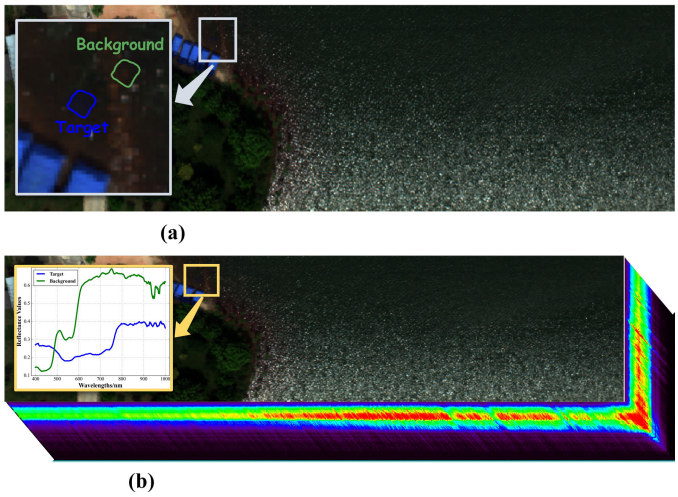


Fig. 1. Illustration of spatial and spectral characteristics in underwater target detection. (a) Conventional optical imagery exhibits limited spatial discriminability between targets and backgrounds. (b) Spectral reflectance curves reveal clear separability between target and background regions.

However, compared with the hyperspectral terrestrial target detection task, HUTD faces a more challenging weak target detection regime. Specifically, underwater propagation effects such as absorption and scattering can severely attenuate target signals and distort their spectral characteristics, making weak targets difficult to characterize reliably. Meanwhile, strong background interference in complex underwater scenes can easily suppress weak target components or absorb them into the dominant background representation, thereby increasing the risk of missed detection. In addition, the inherently low signal-to-noise ratio of underwater HSI makes weak target cues more easily confused with noise-induced responses, leading to increased false alarms. These coupled challenges substantially increase the difficulty of weak underwater tar-

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get detection. To address these challenges, existing studies are commonly divided into prediction-based approaches and separation-based approaches.

Prediction-based methods [13], [14], [15] follow a generative modeling strategy in which key environmental parameters (e.g., water depth, absorption, and scattering coefficients) are first estimated to configure a bathymetric model [9], [10]. A reference spectrum, typically measured in-air or in laboratory conditions, is then propagated through this model to predict its underwater appearance, which is subsequently used for detection with standard hyperspectral techniques such as matched filtering (MF) [16] or spectral angle mapper (SAM) [17]. Representative works include generalized likelihood ratio test (GLRT)-based joint estimation of water properties and target detection [13], physics-driven modeling that incorporates spectral variability and nonlinear reduction [14], and iterative frameworks that refine parameter estimation with spectral and spatial cues [15]. While grounded in physical principles, these methods remain limited in practice. They are highly sensitive to inaccuracies in environmental parameter estimation [18], which is particularly challenging in dynamic aquatic conditions with varying turbidity, salinity, and substrates. In addition, iterative inversion and extensive parameter estimation impose heavy computational costs, restricting scalability and hindering real-time deployment. *In summary*, their performance is fundamentally constrained by dependence on precise environmental parameter estimation.

To overcome this issue, *separation-based methods* [19], [20], [21] adopt a different strategy; instead of estimating environmental parameters, they directly extract target-related information from the observed hyperspectral data. These methods typically employ hyperspectral unmixing [22], [23], where the reference target spectrum is treated as a predefined endmember, and the resulting abundance map is used as the detection output. Prominent studies include a linear unmixing framework that avoids explicit parameter estimation [19], nonlinear extension that better captures spectral variability in complex waters [20], and hybrid approach integrating abundance estimation with spectral fusion and endmember refinement to enhance robustness [21].

Despite these advances, separation-based methods remain constrained by two critical challenges.

1) *Insufficient Preservation of Weak Target Signals:* Most approaches rely on hyperspectral unmixing techniques [22], [23], which were originally designed to extract dominant endmembers corresponding to spatially prevalent materials. In practical remote sensing scenarios, underwater targets usually occupy only a few pixels with limited spatial coverage (as evidenced in Table I), leaving them disadvantaged in the unmixing process. Consequently, strong background components dominate the decomposition, suppressing the abundances of weak target endmembers and often reducing them to zero. This inadequate preservation of weak signals ultimately results in a high rate of missed detections.
2) *Inadequate Incorporation of Imaging-Physics Constraints:* Since hyperspectral unmixing is inherently ill-posed, separation-based methods often lack uniqueness in their solutions. The core objective of unmixing is to determine endmember spectra and corresponding abundances whose linear or nonlinear combinations can reconstruct the observed spectrum. However, multiple combinations may yield equally accurate reconstructions, making the solution space nonunique. Without constraints informed by underwater imaging models [9], [10], optimization tends to emphasize reconstruction fidelity over physical plausibility. As a result, the separated target spectrum may appear numerically consistent with the reference but physically implausible, thereby increasing the likelihood of false alarms.

TABLE I
PIXEL COMPOSITION OF ATR2-HUTD BENCHMARK DATASET

Subdataset	Scene	Target Pixel	Background Pixel
Lake	Scene1	0.0288%	99.9712%
	Scene2	0.0261%	99.9739%
River	Scene1	0.0351%	99.9649%
	Scene2	0.0428%	99.9572%
Sea	Scene1	0.0113%	99.9887%
	Scene2	0.0140%	99.9860%
Average	-	0.0264%	99.9737%

To address the limitations of existing separation-based methods, this work emphasizes the preservation of weak targets through a physically consistent solution guided by the principle that rejection is the best protection. This principle suggests that weak signals can be safeguarded by *explicitly excluding them from the unmixing process*. Accordingly, we design a target-rejection mechanism that enforces spectral separation between extracted endmembers and the known target spectrum, ensuring that the reconstructed image contains no target information. The resulting spectral residuals between the reconstruction and the input are then used to localize potential target regions. However, residuals may also capture noise or irrelevant variations, increasing false alarms. To mitigate this, we further introduce a physics-aware refinement strategy, which evaluates whether candidate pixels with strong residual responses can be physically interpreted as mixtures of the reference target and water medium spectra. Only pixels consistent with the underwater imaging model [9], [10] are retained in the final detection map.

According to these analyses, we propose a physically consistent framework termed rejection-guided residual reasoning network (R^3Net), which integrates a target-rejection mechanism with a physics-aware reasoning strategy to tackle weak target detection in HUTD. The framework first learns endmembers by explicitly maximizing their spectral divergence from the reference target spectrum, ensuring that the extracted endmembers remain distinct from the target and improving the reliability of residual-based detection. Using these endmembers, a background-only reconstruction is generated, and pixel-wise residuals are obtained by subtracting the reconstruction from the original image. High residual responses

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indicate reconstruction failure and thus the potential presence of targets. To suppress false alarms from noise, reconstruction errors, or nontarget weak signals, a physics-aware refinement is applied to the residual map. This step evaluates whether residual responses can be physically interpreted as mixtures of the reference target and water spectra according to the underwater imaging model [9], [10]. Only residuals consistent with this model are retained in the final detection map. Experiments on ATR2-HUTD dataset [24] demonstrate that R^3Net consistently outperforms state-of-the-art (SOTA) approaches, particularly in challenging underwater scenarios. In summary, the main contributions of this work are as follows.

1) We propose a novel HUTD framework grounded in the separation-based paradigm and introduce a new perspective, namely rejection is the best protection. By reformulating HUTD as an unmixing-based signal decomposition problem rather than a direct prior-matching task, the proposed framework better accommodates severe attenuation and spectral distortion in underwater scenes.
2) A target-rejection mechanism is developed to exclude the reference target spectrum during endmember construction, thereby preventing weak target components from being absorbed into the dominant background representation and improving weak target preservation under strong background interference.
3) A physics-aware refinement module is designed to verify the physical consistency of residual responses with the underwater imaging model, thereby suppressing noise-induced false responses and reducing false alarms caused by low signal-to-noise ratio.
4) Extensive experiments on ATR2-HUTD demonstrate that R^3Net achieves superior accuracy, background suppression, and robustness compared with SOTA approaches, particularly in complex underwater environments.

II. Related Work

This section provides the theoretical context of HUTD and outlines the motivations for our approach. We first introduce the fundamentals of hyperspectral underwater imaging (HUI, Section II-A), then review strategies for weak-signal protection (Section II-B), and finally discuss the incorporation of physical imaging mechanisms in HUTD (Section II-C).

A. Hyperspectral Underwater Imaging

Hyperspectral remote sensing has witnessed rapid development in recent years, with representative progress in hyperspectral imaging [25], AI-driven spectral modeling [26], and detection methodologies [27]. Within this broader context, HUI provides the theoretical foundation for HUTD by analyzing how aquatic conditions affect spectral measurements. It characterizes the interactions among incident light, the water column, and submerged objects, where absorption, scattering, and depth-dependent attenuation jointly distort spectral responses and complicate direct interpretation.

A widely adopted principle for modeling these effects is the bathymetric radiative transfer model [9], [10], which expresses the observed spectrum $\boldsymbol{r}(\lambda)$ at wavelength λ as

$$\boldsymbol{r}(\lambda) = \boldsymbol{r}_w(\lambda)\left[1 - e^{-\boldsymbol{K}(\lambda)H}\right] + \boldsymbol{r}_t(\lambda)\, e^{-\boldsymbol{K}(\lambda)H} \tag{1}$$

where $\boldsymbol{r}_w(\lambda)$ is the water-column background spectrum, $\boldsymbol{r}_t(\lambda)$ is the intrinsic target reflectance, $\boldsymbol{K}(\lambda)$ is the total attenuation coefficient, and H is the target depth. This formulation highlights two key properties.

1) The observed spectrum is a depth- and wavelength-dependent mixture of target and water spectra, with weights determined by the exponential attenuation term $e^{-\boldsymbol{K}(\lambda)H}$.
2) The mixture weights follow exponential decay with attenuation and depth, resembling convex unmixing behavior where the contributions of endmembers sum to one and are non-negative.

These properties suggest that underwater spectra can be modeled as structured combinations of known components, motivating the use of spectral unmixing for target detection. Here, the water and target act as spectral endmembers, and their relative contributions can be quantified through abundance estimation as an indicator of target presence. Moreover, the analytical structure of (1) provides a physically interpretable constraint: only spectra consistent with the expected mixing of water and target materials are likely to correspond to true submerged objects. This observation underpins the physics-aware refinement strategy in our framework (Section IV-C), which leverages spectral consistency with the underwater imaging mechanism to suppress residual anomalies and enhance detection reliability.

B. Weak-Signal Protection in HUTD

In real underwater scenes, target pixels typically constitute only a minute fraction of the hyperspectral image. As shown in Table I, in the ATR2-HUTD dataset [24], the target pixels account for merely 0.0264%, indicating that most targets appear as weak signals embedded in complex backgrounds.

Separation-based HUTD methods commonly rely on hyperspectral unmixing, but conventional unmixing emphasizes dominant spectral components and tends to neglect signals with limited spatial presence [28]. This bias severely hinders weak target preservation and degrades practical detection performance. Although weak-signal protection is critical, it has received limited attention. A preliminary attempt [20] incorporated prior-informed initialization and slow-update constraints, where one endmember is initialized with the target spectrum and updated at a reduced learning rate. However, two limitations remain the following.

1) *Underutilization of Prior Supervision:* Due to the scarcity of target pixels, their gradients contribute little to training, and the reduced update rate further diminishes the influence of the target prior, weakening its role in guiding unmixing.
2) *Failure of Signal Preservation:* In complex environments with diverse backgrounds, prolonged optimization gradually distorts the target endmember despite the

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slow-update constraint, undermining prior reliability and sometimes misguiding detection. Our experiments (Section V-B) confirm that the performance of [20] degrades significantly under challenging conditions compared with the simpler settings considered in its original study.

To address these issues, we advocate a new perspective: rejection as protection. Instead of embedding the target prior into unmixing, we explicitly exclude the target spectrum from endmember construction, preventing its absorption into dominant background representations and allowing weak signals to emerge as residuals. To further refine these residuals, we introduce a residual-aware reasoning mechanism with a physics-constrained dynamic masking strategy, which enhances the separation of weak targets from background interference and reduces false activations caused by spectral anomalies or noise.

1) Discussion: Existing methods mainly follow a target-side modeling paradigm, where learning is directly guided by target priors and thus strongly relies on sufficiently informative target cues [20], [28]. However, in nearshore weak underwater target detection, target cues are often extremely weak, unstable, and easily overwhelmed by dominant background components, which limits the effectiveness of target-side modeling and leads to a high risk of missed detection. Different from these methods, our framework adopts a background-centered learning paradigm that organizes learning around background information, which is substantially more abundant, stable, and learnable in real underwater scenes. By combining background-dominant modeling with residual-based target identification, the proposed framework is inherently more suitable for nearshore weak underwater target detection.

C. Physical Imaging Models in HUTD

Incorporating physical constraints has proven effective for enhancing the robustness and interpretability of hyperspectral models [23]. In HUTD, recent studies have integrated underwater imaging mechanisms to better capture spectral characteristics and degradation effects. For instance, SUTDF [15] embeds a depth-aware radiative transfer model into both data synthesis and detection pipelines, TUTDF [29] incorporates an attenuation model within an iterative refinement framework, and NUUTD [20] applies an additive post-nonlinear mixing model to describe water-induced distortions.

Although these approaches employ physics-inspired formulations by modeling spectra as weighted mixtures of target and background components, the associated inverse problems remain inherently ill-posed. Linear or post-nonlinear assumptions allow multiple valid decompositions for the same spectrum, leading to ambiguous interpretations. As a result, such models may yield target-like responses in background regions, increasing false alarms, or fail to recover true target signatures, causing missed detections.

To overcome this limitation, our framework introduces a physical consistency check that verifies whether each candidate pixel conforms to the expected mixture of target and water spectra defined by the underwater imaging mechanism [9], [10]. Only candidates satisfying this constraint are retained, filtering out physically implausible responses and improving detection reliability and interpretability.

1) Discussion: Existing physics-based methods usually rely on explicit physical modeling and often require accurate estimation of environmental parameters to characterize underwater imaging conditions [15], [20], [29]. However, such parameter estimation is often unreliable in nearshore underwater scenes due to complex and highly variable environmental conditions. In contrast, our method does not use the physical model for precise reconstruction or fine-grained parameter estimation. Instead, it distills the physical model into knowledge guidance and embeds it into the optimization and reasoning process, thereby reducing dependence on accurate parameter estimation while improving adaptability to complex nearshore environments.

III. Motivation

Despite recent progress in separation-based methods, real-world HUTD remains hindered by two central challenges: the suppression of weak target signals and the absence of physically grounded constraints. This section analyzes these challenges in detail, clarifying the motivations behind our design choices and laying the foundation for the proposed two-stage underwater weak target detection framework (in Section III-A), which integrates a target-rejection mechanism (in Section III-B) with physics-aware residual reasoning (in Section III-C) to achieve accurate, robust, and interpretable detection.

A. Two-Stage Underwater Weak Target Detection Framework

As illustrated in Fig. 2(a), most existing weak target detection methods follow a *direct target-injection one-stage* paradigm, where the prior target spectrum is explicitly introduced to guide the separation of target-related components. From the perspective of network optimization, however, such prior-guided target-side modeling can be effective only when target-related cues are sufficiently represented during learning [30]. In other words, this paradigm implicitly requires the target signal itself to provide sufficiently informative and stable evidence for subsequent target modeling. Such an assumption is often violated in underwater weak target detection, where target responses are typically extremely weak, spatially sparse, and easily submerged by dominant background components. As a result, directly organizing detection around target-side modeling is inherently mismatched with the weak-target nature of underwater scenes.

To overcome this limitation, we reformulate the original direct target-injection one-stage process into an *indirect target-rejection two-stage* framework, as shown in Fig. 2(b). The key idea is that, instead of forcing weak targets to be explicitly modeled at the beginning, the proposed framework first organizes learning around the more dominant and learnable background information, and then identifies target-related responses from the residual domain. In this way, the detection pipeline is no longer built upon scarce and unstable target cues, but upon the background-dominant structure that is more consistent with the actual data characteristics of underwater scenes.

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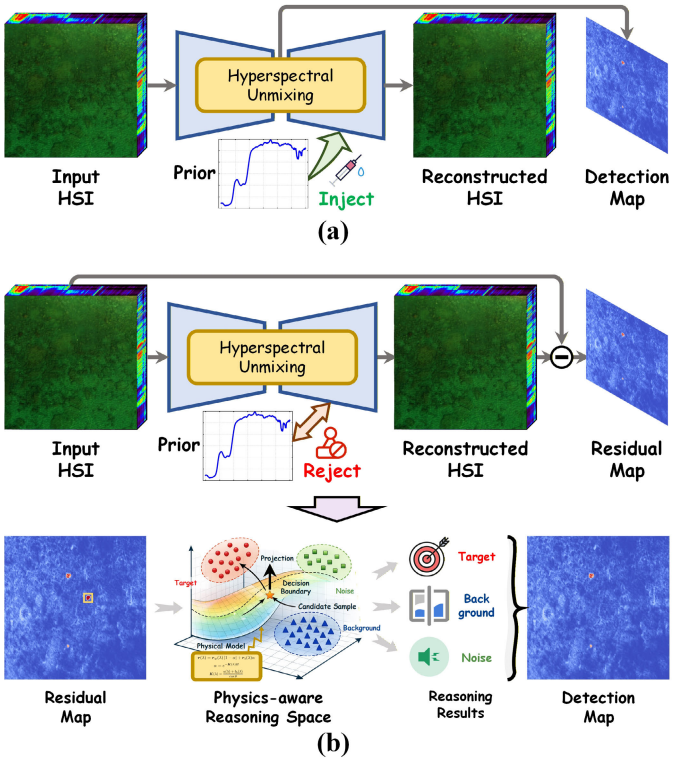


Fig. 2. Comparison of different weak-signal detection framework in HUTD. (a) One-stage target-injection framework [20]. (b) Two-stage target-rejection framework proposed in this work.

Specifically, in the first stage, the framework performs background-oriented reconstruction by explicitly excluding the target prior from the reconstruction process. This prevents weak target cues from being absorbed into dominant background modeling, while preserving target-related information that cannot be explained by the background in the residual domain. Therefore, the first stage effectively alleviates the missed-detection risk caused by strong background interference. In the second stage, the framework further verifies the physical consistency of residual responses with the underwater imaging model, so that noise-induced and physically implausible responses can be suppressed. As a result, the proposed two-stage framework not only improves weak target preservation but also reduces false alarms under low signal-to-noise ratio conditions.

B. Rejection-Based Weak-Signal Protection

Fig. 2(a) shows the target-injection strategy proposed in the pioneering study [20], where the reference target spectrum is fixed as a predefined endmember in the unmixing model. Based on unmixing theory [23], this endmember contributes only when applied to pixels with nonzero target abundance. As a result, its activation depends on the limited target pixels, making the strategy effective only when sufficient target samples are available [23]. This creates an inherent contradiction: the strategy is intended to preserve weak signals under target scarcity but functions effectively only when targets are abundant. Such self-contradictory behavior fundamentally limits its practicality in real-world applications.

The failure of target injection arises from pixel scarcity, suggesting that weak-signal protection should instead *exploit richer background information*, which dominates the HSI. Based on this insight, we propose a new design principle: rejection is the best protection. As shown in Fig. 2(b), we develop a target-rejection pipeline that excludes the reference spectrum during endmember construction, ensuring reconstructed HSIs are background-only. Subtracting this reconstruction from the input produces a residual map that highlights pixels inconsistent with background spectra and serves as a reliable cue for detection.

This rejection-reconstruction design offers two advantages.

1) *Improved Weak-Signal Preservation:* By enforcing reconstruction solely from background information, target responses are isolated in the residual map rather than absorbed into background endmembers. This yields clearer separation and avoids reliance on target pixel availability, aligning naturally with weak-signal detection requirements.
2) *Enhanced Convergence:* Target pixels often behave as spectral outliers that destabilize optimization. Excluding them enables unmixing to focus on background spectra, leading to more stable updates and faster convergence to better solutions.

Nevertheless, residual maps may still contain false activations from spectral noise, incomplete background modeling, or nontarget weak signals. To mitigate these issues, we introduce a physics-aware residual reasoning strategy, detailed in Section III-C.

C. Physics-Aware Residual Reasoning

The target-rejection module produces a residual map $\mathcal{D}_r$, from which candidate pixels $\{\boldsymbol{r}_c^i\}_{i=1}^N$ are extracted by thresholding with τ_1

$$\left\{\boldsymbol{r}_c^i\right\}_{i=1}^N = \left\{\boldsymbol{r}_p \;\middle|\; \mathcal{D}_r(p) > \tau_1,\ p \in \Omega\right\} \tag{2}$$

where $\boldsymbol{r}_p \in \mathbb{R}^L$ denotes the spectrum of pixel p and Ω is the set of all pixel locations. According to the underwater imaging mechanism in (1), a true target pixel should approximately follow a two-endmember mixing model:

$$\boldsymbol{r}_c^i \;\approx\; \alpha_i \boldsymbol{r}_t + (1-\alpha_i)\,\boldsymbol{r}_w, \quad \alpha_i \in [0,1] \tag{3}$$

where $\boldsymbol{r}_t$ and $\boldsymbol{r}_w$ are the target and water spectra, and α_i is the target abundance. The background spectrum $\boldsymbol{r}_w$ can be robustly estimated using classical unmixing methods [19], [20].

Building on this model, each candidate pixel can be evaluated by how well it conforms to the expected target–water mixture. To this end, we define a reliability score as the negative reconstruction error under (3)

$$\mathcal{R}\left(\boldsymbol{r}_c^i\right) \;=\; -\min_{\alpha_i \in [0,1]} \left\| \boldsymbol{r}_c^i - \left(\alpha_i \boldsymbol{r}_t + (1-\alpha_i)\,\boldsymbol{r}_w\right) \right\|_2 \tag{4}$$

where larger $\mathcal{R}$ indicates higher physical consistency. The optimization is a 1-D bounded least-squares problem solvable efficiently via linear programming [31]. Candidate pixels with $\mathcal{R} > \tau_2$ are retained to form the final detection result

$$\mathcal{D}_f = \left\{\boldsymbol{r}_c^i \;\middle|\; \mathcal{R}\left(\boldsymbol{r}_c^i\right) > \tau_2,\ i = 1, \ldots, N\right\}. \tag{5}$$

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This physics-aware reasoning enforces physical plausibility but suffers from strong dependence on thresholds τ_1 and τ_2, reducing generalization across scenes. To overcome this issue, we incorporate a dynamic masking mechanism that transforms local thresholding into a global decision process guided by the distribution of residuals, thereby enhancing robustness and eliminating manual tuning (see Section IV-C).

Finally, although (4) could in principle be applied to all pixels, such exhaustive evaluation is computationally prohibitive. Moreover, as shown in Section V-C1, this brute-force strategy not only incurs heavy computational cost but also degrades detection performance. To overcome these issues, the residual map generated in Section III-B plays a dual role: it preserves weak target signals while simultaneously narrowing the reasoning space. This design allows computational resources to be concentrated on the most promising candidate regions, leading to solutions that are closer to the optimum in both accuracy and efficiency. Consequently, the proposed physics-aware residual reasoning strategy achieves a balance among detection reliability, physical interpretability, and computational tractability.

D. Main Innovations

Based on the above motivation and design logic, the proposed framework introduces three core methodological innovations for weak underwater target detection.

1) *A Two-Stage Weak Target Detection Paradigm:* Unlike conventional one-stage target-side modeling pipelines, the proposed framework reformulates weak underwater target detection as a two-stage process consisting of target-rejection-guided reconstruction and residual reasoning. Instead of detecting targets through direct target modeling, it first models the dominant background and then identifies target-related responses from the residual domain. This paradigm shift is better aligned with the weak-target nature of underwater scenes and effectively alleviates missed detections caused by strong background interference.
2) *A Target-Rejection-Based Weak Target Preservation Mechanism:* A target-rejection mechanism is introduced into the reconstruction process to explicitly exclude the reference target spectrum from endmember construction. In this way, weak target cues are prevented from being absorbed into the dominant background representation and can be preserved in the residual domain. Compared with conventional target-side protection strategies, this mechanism is more suitable for the scarce and unstable characteristics of weak underwater targets.
3) *A Physics-Aware Residual Verification Strategy:* A physics-aware reasoning module is further designed to verify the physical consistency of residual responses with the underwater imaging model. By filtering out noise-induced and physically implausible responses, this strategy effectively suppresses false alarms caused by low signal-to-noise ratio while improving the physical reliability of detection results.

IV. Method

This section first presents the overall framework of R^{3}Net to provide a comprehensive view of its design and processing pipeline. We then elaborate on each component, explaining its role within the framework, the principles motivating its design, and the key implementation details.

A. Overall Framework

Building on the design principles in Section III, we propose R^{3}Net, a separation-based framework tailored for weak target detection in HUTD. As illustrated in Fig. 3, R^{3}Net consists of two sequential stages: *rejection-guided spectral reconstruction* (Section IV-B) and *physics-aware residual reasoning* (Section IV-C).

In the first stage, R^{3}Net reconstructs a background-only hyperspectral image by explicitly excluding the reference target spectrum from the endmember set. The input HSI is processed by a 3-D convolutional neural network (CNN) encoder and fully connected layers to estimate pixel-wise abundances under a Softmax-normalized simplex constraint. A decoder then combines the learned endmembers with the estimated abundances to generate the reconstruction. To ensure compact and discriminative background representations, sparsity, repulsion, and fidelity constraints are imposed. The target-suppressed reconstruction is subtracted from the input to form a residual map that highlights potential targets.

In the second stage, this residual map is refined using physics-aware reasoning. A learnable spatial mask is initialized from the residual and optimized with an objective derived from the underwater radiative transfer model [see (1)], enforcing spectral consistency with target–water mixtures and collapse prevention through an ℓ_2 penalty. The optimized mask yields accurate detections without manual thresholding.

By integrating rejection-guided reconstruction with physics-aware reasoning, R^{3}Net achieves robust and interpretable underwater target detection, particularly under conditions of weak targets and strong background interference.

B. Rejection-Guided Spectral Reconstruction

This section introduces rejection-guided spectral reconstruction, designed to highlight underwater targets with weak spectral responses. It consists of two sequential steps: *rejection-guided unmixing*, which generates a background-only reconstruction by structurally excluding the target spectrum, and *reconstruction-based detection*, which derives residual maps from the difference between the input and reconstruction to reveal potential targets.

1) Rejection-Guided Unmixing: Given an input HSI $\boldsymbol{X} \in \mathbb{R}^{H\times W\times B}$ and a reference target spectrum $\boldsymbol{r}_t \in \mathbb{R}^B$, this step decomposes each pixel into background-only endmembers while explicitly excluding $\boldsymbol{r}_t$. An autoencoder unmixing network is employed, combining a target-rejection mechanism with auxiliary regularizations.

As illustrated in Fig. 3, an encoder first extracts spatial–spectral features through L_1 layers of 3-D convolutions with ReLU activations, followed by L_2 fully connected layers with Tanh activations for nonlinear abstraction. A SoftMax

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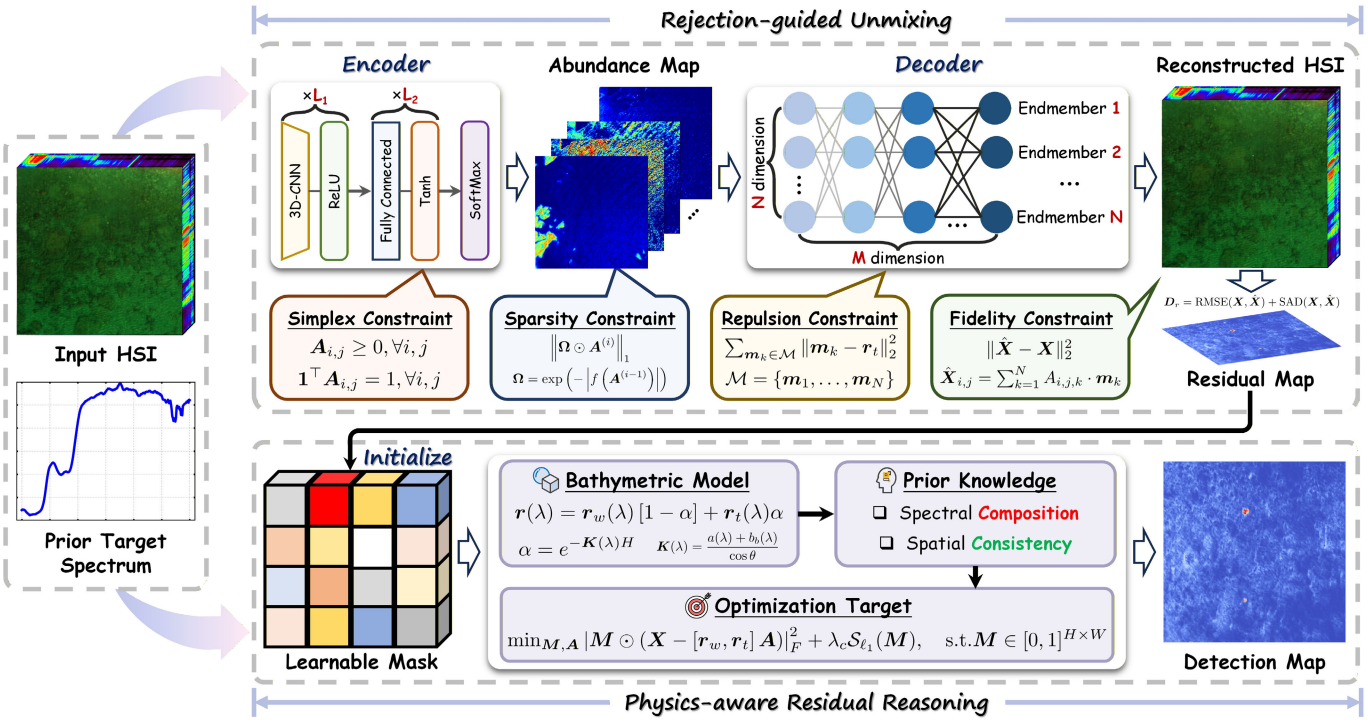


Fig. 3. Overview of R^{3}Net. R^{3}Net consists of two sequential modules: 1) a rejection-guided spectral reconstruction module, which reconstructs a background-only hyperspectral image by structurally excluding the reference target spectrum; and 2) a physics-aware residual reasoning module, which refines the residual map based on spectral consistency with the underwater imaging model. This two-stage design enables robust and physically interpretable detection of weak underwater targets.

layer then produces abundance vectors that satisfy nonnegativity and sum-to-one constraints [32], yielding the abundance map $\boldsymbol{A} \in \mathbb{R}^{H\times W\times N}$, where N is the number of endmembers.

To reflect the intuition that underwater pixels usually consist of only a few dominant materials, we impose an attention-based sparsity constraint

$$\mathcal{L}_{\text{sparsity}} = \left\| \exp\left(-\left|f\left(\boldsymbol{A}^{(i-1)}\right)\right|\right) \odot \boldsymbol{A}^{(i)} \right\|_1 \tag{6}$$

where $\boldsymbol{A}^{(i)}$ is the abundance map at iteration i, $f(\cdot)$ is a spatial filter, $\|\cdot\|_1$ is the ℓ_1 norm, and $\odot$ is the element-wise product. Here, the previous abundance $\boldsymbol{A}^{(i-1)}$ is used to derive the attention weights $\boldsymbol{W}^{(i)} = \exp(-|f(\boldsymbol{A}^{(i-1)})|)$ so that (6) realizes a reweighted ℓ_1 penalty. Salient and spatially coherent entries are penalized less, while weak or diffuse ones are suppressed more, yielding fine-grained and stable sparsity.

In parallel, the model learns a set of background endmembers $\boldsymbol{E} = \{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_N\}$, initialized using Robust-OSP [28]. To prevent contamination by target information, a repulsion loss enforces spectral separation between each $\boldsymbol{e}_n$ and $\boldsymbol{r}_t$

$$\mathcal{L}_{\text{repulse}} = \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{e}_n - \boldsymbol{r}_t\|_2^2 \tag{7}$$

where $\|\cdot\|_2$ is the ℓ_2 norm. The reconstructed background-only HSI $\hat{\boldsymbol{X}}$ is obtained as

$$\hat{\boldsymbol{X}}_{i,j} = \sum_{n=1}^{N} A_{i,j,n} \cdot \boldsymbol{e}_n \tag{8}$$

with fidelity enforced by

$$\mathcal{L}_{\text{rec}} = \left\| \hat{\boldsymbol{X}} - \boldsymbol{X} \right\|_2^2. \tag{9}$$

The overall loss integrates reconstruction, sparsity, and repulsion

$$\mathcal{L}_{\text{unmix}} = \mathcal{L}_{\text{rec}} + \lambda_s \mathcal{L}_{\text{sparsity}} + \lambda_r \mathcal{L}_{\text{repulse}} \tag{10}$$

where $\lambda_s, \lambda_r > 0$ balance the regularizations. This design ensures that the learned endmembers are compact, discriminative, and structurally free of target spectra, providing a reliable background reconstruction for residual-based detection.

2) Reconstruction-Based Detection: After rejection-guided unmixing, the input HSI $\boldsymbol{X}$ is approximated by a background-only reconstruction $\hat{\boldsymbol{X}}$. Since the target spectrum is structurally excluded, discrepancies between $\boldsymbol{X}$ and $\hat{\boldsymbol{X}}$ naturally indicate potential target presence.

To quantify these discrepancies, we construct a residual-based detection map (RDM) $\boldsymbol{D}_r \in \mathbb{R}^{H\times W}$ by fusing two complementary metrics: the normalized ℓ_2 reconstruction error and the normalized spectral angle distance. The detection score for each pixel (i, j) is defined as

$$\boldsymbol{D}_r(i,j) = \frac{1}{2}\left[\frac{\left\|\boldsymbol{X}_{i,j,:} - \hat{\boldsymbol{X}}_{i,j,:}\right\|_2}{\max_{(p,q)}\left\|\boldsymbol{X}_{p,q,:} - \hat{\boldsymbol{X}}_{p,q,:}\right\|_2} + \frac{1}{\pi}\arccos\left(\frac{\boldsymbol{X}_{i,j,:}^{\top}\hat{\boldsymbol{X}}_{i,j,:}}{\left\|\boldsymbol{X}_{i,j,:}\right\|_2\left\|\hat{\boldsymbol{X}}_{i,j,:}\right\|_2}\right)\right] \tag{11}$$

where $\boldsymbol{X}_{i,j,:}$ and $\hat{\boldsymbol{X}}_{i,j,:}$ are the observed and reconstructed spectra at (i, j) and $\max_{(p,q)}\|\boldsymbol{X}_{p,q,:} - \hat{\boldsymbol{X}}_{p,q,:}\|_2$ represents the global maximum residual calculated across the entire image.

In (11), $\boldsymbol{D}_r(i, j)$ is designed to measure how poorly pixel (i, j) can be explained by the reconstructed background spectrum. Specifically, the first term computes the normalized

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ℓ_2 reconstruction error between $\boldsymbol{X}_{i,j,:}$ and $\hat{\boldsymbol{X}}_{i,j,:}$, which reflects the magnitude discrepancy between the observed and reconstructed spectra. The second term computes the normalized spectral angle distance, which captures their spectral-shape discrepancy. These two terms are complementary: the former is sensitive to overall reconstruction error, whereas the latter is sensitive to directional spectral deviation even when the magnitude difference is limited. To make them comparable before fusion, the ℓ_2 error is normalized by the global maximum residual over the image, and the spectral angle is normalized by π, so that both terms are mapped into $[0, 1]$. Their equal-weight average is then used to construct a pixel-wise background-unexplainability score. Therefore, a larger $\boldsymbol{D}_r(i, j)$ indicates that the corresponding pixel deviates more strongly from the reconstructed background and is thus more likely to contain target-related information. The resulting map $\boldsymbol{D}_r$ provides a coarse localization of potential targets and serves as essential input for the physics-aware refinement module in Section IV-C.

C. Physics-Aware Residual Reasoning

Although the residual-based map $\boldsymbol{D}_r$ provides coarse localization of potential underwater targets, it inevitably contains false positives due to background noise, imperfect reconstruction, and other nontarget weak signals. This issue is particularly critical in weak-signal regimes, where true targets have low spectral contrast and can be confused with natural background variations. To enhance precision and suppress unreliable activations, we introduce a physics-aware residual reasoning strategy that evaluates the spectral consistency of candidate pixels with the underwater imaging mechanism. By incorporating constraints derived from radiative transfer theory, this module enables a principled refinement process and improves detection reliability under complex conditions.

1) Residual Reasoning Formulation: The goal is to verify whether an observed spectrum can be physically explained as a linear mixture of the water spectrum $\boldsymbol{r}_w$ and the target spectrum $\boldsymbol{r}_t$. However, as discussed in Section III-C, direct pixel-wise thresholding is sensitive to scene-dependent parameters.

To mitigate this sensitivity, we reformulate the process as a global optimization problem that selects a spatially coherent subset of pixels whose spectra conform to the target–water mixing model: pixels with large reconstruction errors are penalized, while those with small errors are favored. This replaces rigid per-pixel thresholding with a flexible and computationally efficient optimization framework.

Specifically, we introduce a learnable spatial mask $\boldsymbol{M} \in [0, 1]^{H\times W}$ that assigns pixel-wise target confidence, together with an abundance map $\boldsymbol{H} \in \mathbb{R}^{H\times W\times 2}$ describing the water–target mixture. Minimizing the reconstruction loss alone can cause $\boldsymbol{M}$ to collapse to zeros to avoid penalties. We therefore add an ℓ_2 regularizer on $\boldsymbol{M}$ to prevent collapse and to preserve informative pixels, which empirically improves recall. The complete formulation is

$$\begin{aligned} \min_{\boldsymbol{M},\boldsymbol{H}} \quad & \left\| \boldsymbol{M} \odot (\boldsymbol{X} - [\boldsymbol{r}_w, \boldsymbol{r}_t]\, \boldsymbol{H}) \right\|_F \;+\; \lambda_c \|\boldsymbol{M}\|_2 \\ \text{s.t.} \quad & \boldsymbol{M} \in [0, 1]^{H\times W}, \quad \boldsymbol{H} \geq 0,\ \mathbf{1}^\top \boldsymbol{H}_{i,j} = 1\ \forall i, j \end{aligned} \tag{12}$$

where $\odot$ is the element-wise product, $\|\cdot\|_F$ is the Frobenius norm, and $\lambda_c > 0$ balances reconstruction fidelity and mask stability.

2) Solution and Optimization: For tractability, we reshape the optimization problem into a compact matrix form

$$\begin{aligned} \min_{\tilde{\boldsymbol{M}},\tilde{\boldsymbol{H}}} \quad & \left\| \tilde{\boldsymbol{M}} \odot \left(\tilde{\boldsymbol{X}} - \boldsymbol{R}\tilde{\boldsymbol{H}}\right) \right\|_F \;+\; \lambda_c \left\| \tilde{\boldsymbol{M}} \right\|_2 \\ \text{s.t.} \quad & \tilde{\boldsymbol{M}} \in [0, 1]^{1\times Q}, \quad \tilde{\boldsymbol{H}} \geq 0,\ \mathbf{1}^\top \tilde{\boldsymbol{H}} = \mathbf{1}^\top \end{aligned} \tag{13}$$

where $\tilde{\boldsymbol{X}} \in \mathbb{R}^{B\times Q}$ stacks $Q = H \times W$ pixels column-wise, $\boldsymbol{R} = [\boldsymbol{r}_w, \boldsymbol{r}_t] \in \mathbb{R}^{B\times 2}$, $\tilde{\boldsymbol{H}} \in \mathbb{R}^{2\times Q}$ is the abundance matrix, and $\tilde{\boldsymbol{M}} \in \mathbb{R}^{1\times Q}$ is the reshaped mask. The constraint $\mathbf{1}^\top \tilde{\boldsymbol{H}} = \mathbf{1}^\top$ enforces the simplex condition. To embed the sum-to-one constraint implicitly, we adopt an augmentation trick

$$\boldsymbol{R} \leftarrow \begin{bmatrix} \boldsymbol{R} \\ \alpha \mathbf{1}_2^\top \end{bmatrix}, \quad \tilde{\boldsymbol{X}} \leftarrow \begin{bmatrix} \tilde{\boldsymbol{X}} \\ \alpha \mathbf{1}_Q^\top \end{bmatrix} \tag{14}$$

where $\alpha > 0$ controls the constraint strength and $\mathbf{1}_2^\top \in \mathbb{R}^{1\times 2}$ and $\mathbf{1}_Q^\top \in \mathbb{R}^{1\times Q}$ are all-one vectors. This avoids explicit equality constraints during optimization.

Since (13) couples $\tilde{\boldsymbol{M}}$ and $\tilde{\boldsymbol{H}}$ bilinearly, we adopt alternating optimization: fixing $\tilde{\boldsymbol{M}}$ to update $\tilde{\boldsymbol{H}}$ and then fixing $\tilde{\boldsymbol{H}}$ to update $\tilde{\boldsymbol{M}}$. Taking the mask update as example, with $\tilde{\boldsymbol{H}}^{(t)}$ fixed, the subproblem reduces to

$$\min_{\tilde{\boldsymbol{M}}\in[0,1]^{1\times Q}} \left\| \left(\tilde{\boldsymbol{X}} - \boldsymbol{R}\tilde{\boldsymbol{H}}^{(t)}\right) \odot \tilde{\boldsymbol{M}} \right\|_F + \lambda_c \left\| \tilde{\boldsymbol{M}} \right\|_2. \tag{15}$$

Since the subproblem objective $F(\tilde{\boldsymbol{M}}, \tilde{\boldsymbol{H}}^{(t)})$ in (15) is convex regarding $\tilde{\boldsymbol{M}}$ with Lipschitz-continuous gradient, we adopt Nesterov's optimal gradient method (OGM) for accelerated convergence. At inner iteration k, the original objective is locally approximated by a strongly convex surrogate around the extrapolated point $\boldsymbol{S}_{k-1}$

$$\begin{aligned} \tilde{\boldsymbol{M}}_k = \arg\min_{\tilde{\boldsymbol{M}}\in[0,1]^{1\times Q}} \Bigg\{ & F\left(\boldsymbol{S}_{k-1}, \tilde{\boldsymbol{H}}^{(t)}\right) \\ & + \left\langle \nabla_{\tilde{\boldsymbol{M}}} F\left(\boldsymbol{S}_{k-1}, \tilde{\boldsymbol{H}}^{(t)}\right), \tilde{\boldsymbol{M}} - \boldsymbol{S}_{k-1} \right\rangle \\ & + \frac{L_C}{2} \left\| \tilde{\boldsymbol{M}} - \boldsymbol{S}_{k-1} \right\|_F \Bigg\} \end{aligned} \tag{16}$$

where $\nabla_{\tilde{\boldsymbol{M}}} F(\boldsymbol{S}_{k-1}, \tilde{\boldsymbol{H}}^{(t)})$ denotes the gradient of F with respect to $\tilde{\boldsymbol{M}}$, evaluated at $\boldsymbol{S}_{k-1}$. The parameter L_C is the Lipschitz constant that guarantees convergence, defined as

$$L_C = \left\| \nabla^2_{\tilde{\boldsymbol{M}}} F\left(\boldsymbol{S}_{k-1}, \tilde{\boldsymbol{H}}^{(t)}\right) \right\|_2. \tag{17}$$

In (16), the first two terms define a first-order Taylor expansion of the objective at $\boldsymbol{S}_{k-1}$, while the third term acts as a proximal regularizer that penalizes deviations from the current iterate, ensuring stability and convergence. This yields a strongly convex surrogate function that admits a closed-form update, leading to the projected gradient step

$$\tilde{\boldsymbol{M}}_k = \mathcal{P}_{[0,1]}\left(\boldsymbol{S}_{k-1} - \frac{1}{L_C} \nabla_{\tilde{\boldsymbol{M}}} F\left(\boldsymbol{S}_{k-1}, \tilde{\boldsymbol{H}}^{(t)}\right) \right) \tag{18}$$

where $\mathcal{P}_{[0,1]}(\cdot)$ projects onto the interval $[0, 1]$. To accelerate convergence, the search point is updated with Nesterov extrapolation

$$\boldsymbol{S}_k = \tilde{\boldsymbol{M}}_k + \frac{\gamma_{k-1} - 1}{\gamma_k} \left(\tilde{\boldsymbol{M}}_k - \tilde{\boldsymbol{M}}_{k-1}\right) \tag{19}$$

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where the extrapolation factor follows the recurrence:

$$\gamma_k = \frac{1 + \sqrt{1 + 4\gamma_{k-1}^2}}{2}. \tag{20}$$

Algorithm 1 Subproblem Optimization

Input: Current mask $\tilde{\boldsymbol{M}}^{(t)}$; fixed abundances $\tilde{\boldsymbol{H}}^{(t)}$
Output: Updated mask $\tilde{\boldsymbol{M}}^{(t+1)}$
Initialization: $\tilde{\boldsymbol{M}}_0 = \tilde{\boldsymbol{M}}^{(t)}$, $\boldsymbol{S}_0 = \boldsymbol{S}^{(t)}$, $\gamma_0 = 1$, $k = 0$
Set L_C using Eq. (17) ▷Compute Lipschitz constant
repeat
  Update $\tilde{\boldsymbol{M}}_k$ using Eq. (18) ▷Perform projected gradient descent
  Update $\boldsymbol{S}_k$ using Eq. (19) ▷Perform momentum extrapolation
  Update γ_k using Eq. (20) ▷Update extrapolation factor
  $k \leftarrow k + 1$ ▷Increment iteration counter
until Criterion in Eq. (21) holds ▷Check convergence
Update $\tilde{\boldsymbol{M}}^{(t+1)} \leftarrow \tilde{\boldsymbol{M}}_k$ ▷Assign final solution
return $\tilde{\boldsymbol{M}}^{(t+1)}$

The iterative procedure is summarized in Algorithm 1, with convergence determined by the projected gradient norm

$$\left\| \nabla^P_{\tilde{\boldsymbol{M}}} F\left(\tilde{\boldsymbol{M}}_k, \tilde{\boldsymbol{H}}^{(t)}\right) \right\|_F \le 10^{-3} \left\| \nabla^P_{\tilde{\boldsymbol{M}}} F\left(\tilde{\boldsymbol{M}}^{(1)}, \tilde{\boldsymbol{H}}^{(1)}\right) \right\|_F \tag{21}$$

where $\nabla^P_{\tilde{\boldsymbol{M}}} F(\cdot)$ denotes the projected gradient operator as

$$\left(\nabla^P_{\tilde{\boldsymbol{M}}} F\left(\tilde{\boldsymbol{M}}_k, \tilde{\boldsymbol{H}}^{(t)}\right)\right)_{ij} = \begin{cases} \left(\nabla_{\tilde{\boldsymbol{M}}} F\left(\tilde{\boldsymbol{M}}_k, \tilde{\boldsymbol{H}}^{(t)}\right)\right)_{ij}, & \text{if } \left(\tilde{\boldsymbol{M}}_k\right)_{ij} > 0 \\ \min\left\{0, \left(\nabla_{\tilde{\boldsymbol{M}}} F\left(\tilde{\boldsymbol{M}}_k, \tilde{\boldsymbol{H}}^{(t)}\right)\right)_{ij}\right\}, & \text{if } \left(\tilde{\boldsymbol{M}}_k\right)_{ij} = 0. \end{cases} \tag{22}$$

Algorithm 2 Physics-Aware Residual Reasoning

Input: Reshaped Hyperspectral image $\tilde{\boldsymbol{X}}$; reference target spectrum $\boldsymbol{r}_t$; water spectrum $\boldsymbol{r}_w$; residual map $\boldsymbol{D}_r$; iteration limit T; hyperparameters λ_c, α
Output: Detection map $\boldsymbol{D}$
Initialize: Set $\tilde{\boldsymbol{M}}^{(0)} \leftarrow \boldsymbol{D}_r$
return $\boldsymbol{D} \leftarrow \tilde{\boldsymbol{M}}^{(T)}$ ▷Return final detection map

The abundance matrix $\tilde{\boldsymbol{H}}$ is updated with the same alternating strategy as in Algorithm 1. Through T iterations, the mask $\tilde{\boldsymbol{M}}$ and abundance matrix $\tilde{\boldsymbol{H}}$ are refined in turn until convergence. The final mask $\tilde{\boldsymbol{M}}^{(T)}$ serves as the detection output, providing pixel-wise confidence scores in $[0, 1]$. The overall optimization process of the physics-aware residual reasoning module is summarized in Algorithm 2.

D. Initialization

In this module, the only variable requiring initialization is the spatial confidence mask $\tilde{\boldsymbol{M}}$. Because our goal is to refine residual-based detections rather than discover targets from scratch, we initialize $\tilde{\boldsymbol{M}}^{(0)}$ with the residual map $\boldsymbol{D}_r$ (Section IV-B2), which highlights spectra deviating from the reconstructed background and serves as a data-driven prior.

This initialization is critical for stability and effectiveness. The reasoning problem is nonconvex due to the bilinear coupling between mask and abundances, and an informative initialization steers optimization toward desirable minima. Empirically, uniform or random initialization often yields degenerate solutions (detecting no targets or labeling the entire scene as target), whereas the residual-based initialization consistently improves convergence and accuracy (see Section V-C1).

Beyond the weak-signal preservation and target–background separation, $\boldsymbol{D}_r$ thus provides a physically interpretable feature that improves robustness in the subsequent reasoning stage. In summary, rejection-guided reconstruction not only suppresses target responses but also produces a reliable initialization that ensures stable convergence and improved detection results.

V. Experiments

A. Experimental Setup

1) Datasets: To evaluate the performance of the proposed framework, we utilize the ATR2-HUTD [24] benchmark.[1] This large-scale dataset consists of HSI acquired by a UAV platform across six distinct real-world underwater scenes. Data acquisition was performed using a Nano-Hyperspec sensor from Headwall Photonics mounted on a DJI Matrice 300 RTK UAV. The collection sites include Qianlu Lake in Liuyang City, Xiangjiang River in Changsha City, and the coastal waters of Sanya City, China. A key feature of this benchmark is the diversity of the underwater environments, which provides varying degrees of turbidity, salinity, and spectral attenuation. Such diversity enables a comprehensive analysis of the robustness of the detection model under complex optical conditions. Furthermore, the dataset includes various submerged targets, such as the *Blue Metal Plate*, *Black Plastic Plate*, and *Yellow Wooden Board*. These objects exhibit distinct visual and spectral properties, allowing for the assessment of detection generalization based on material and color. The depth of the targets ranges from 0.64 to 2.74 m, which facilitates the evaluation of adaptability to different levels of submersion.

2) Evaluation Metrics: To assess detection performance, we employ both qualitative and quantitative metrics, including detection maps, receiver operating characteristic (ROC) analysis, target–background separability (TBS) maps, and area under curve (AUC)-derived indicators.

3) Comparison Methods: We compare R^{3}Net with a broad set of hyperspectral target detection (HTD) and HUTD baselines. The benchmark includes two classical HTD techniques (MF [16] and SAM [17]), four prediction-driven HUTD methods (TUTDF [29], HUCLNet [33], and PCL-HUTD [34]), and four separation-based HUTD frameworks (UTD-Net [19], NUN-UTD [20], HUTDNet [21], and SVHAE [35]). For HUCLNet [33], the more competitive version with SAM detector is employed for comparison. Besides the full R^{3}Net, we also report the RDM, derived from the rejection-guided

[1]The benchmark is available at https://github.com/qjh1996/HUTD

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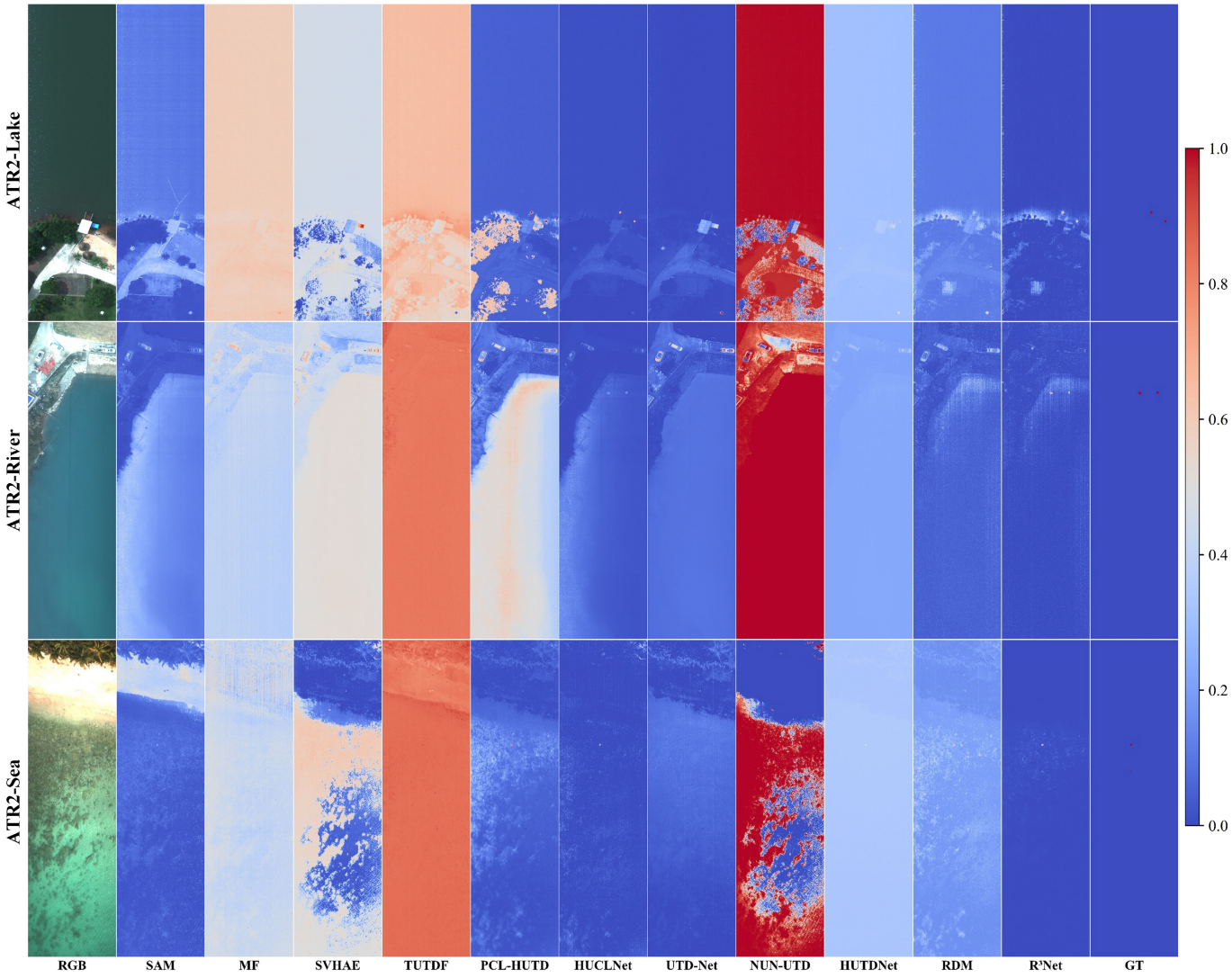


Fig. 4. Detection map comparison on two scenes from the ATR2-HUTD-Lake subset. Each column corresponds to a different method, and each row shows results for a specific scene.

reconstruction module, to validate the core idea that rejection is the best protection for weak target detection.

4) Implementation Details: The rejection-guided spectral reconstruction module employs five 3-D convolutional layers with kernel size $3 \times 3 \times 5$, spatial stride 1, and spectral stride 2. Spatial padding of 1 preserves resolution, while spectral padding of 2 maintains alignment. Two fully connected layers follow to estimate abundance maps with a balance of complexity and efficiency. For the decoder, the number of endmembers is set to $N = 6, 12, 8$ for the *Lake*, *River*, and *Sea* subsets. The attention-based sparsity constraint [see (6)] uses a mean filter $f(\cdot)$, and the unmixing loss [see (10)] adopts $\lambda_s = 0.001$ and $\lambda_r = 10$. In the physics-aware residual reasoning module, regularization weights $\lambda_c = 0.005$ and $\alpha = 5$ are used, with iteration number $T = 100$ to ensure convergence. The impact of these hyperparameters is analyzed in Section V-D. All experiments are conducted on a workstation with seven NVIDIA A6000 GPUs, an AMD Ryzen 5995WX CPU, 128-GB RAM, and Ubuntu 22.04.

B. Main Results

1) Detection Map: Pixel-level detection maps from three representative scenes are shown in Fig. 4. Classical HTD baselines, including SAM and MF, produce relatively smooth response distributions with limited contrast between target regions and surrounding water. Their responses tend to extend over large shallow-water or seabed areas, indicating a strong sensitivity to background variations and resulting in widespread activations that are inconsistent with the sparse target distributions indicated by the ground truth.

Prediction-driven HUTD methods generate more pronounced responses around potential targets, while exhibiting distinct response characteristics. TUTDF yields comparatively more concentrated response patterns, although noticeable diffusion remains in background areas across different scenes. PCL-HUTD and HUCLNet increase local response contrast around target-like structures; however, scattered responses that do not correspond to ground-truth target locations can still be observed, particularly in turbid or high-reflectance water regions. SVHAE exhibits poor target localization and weak

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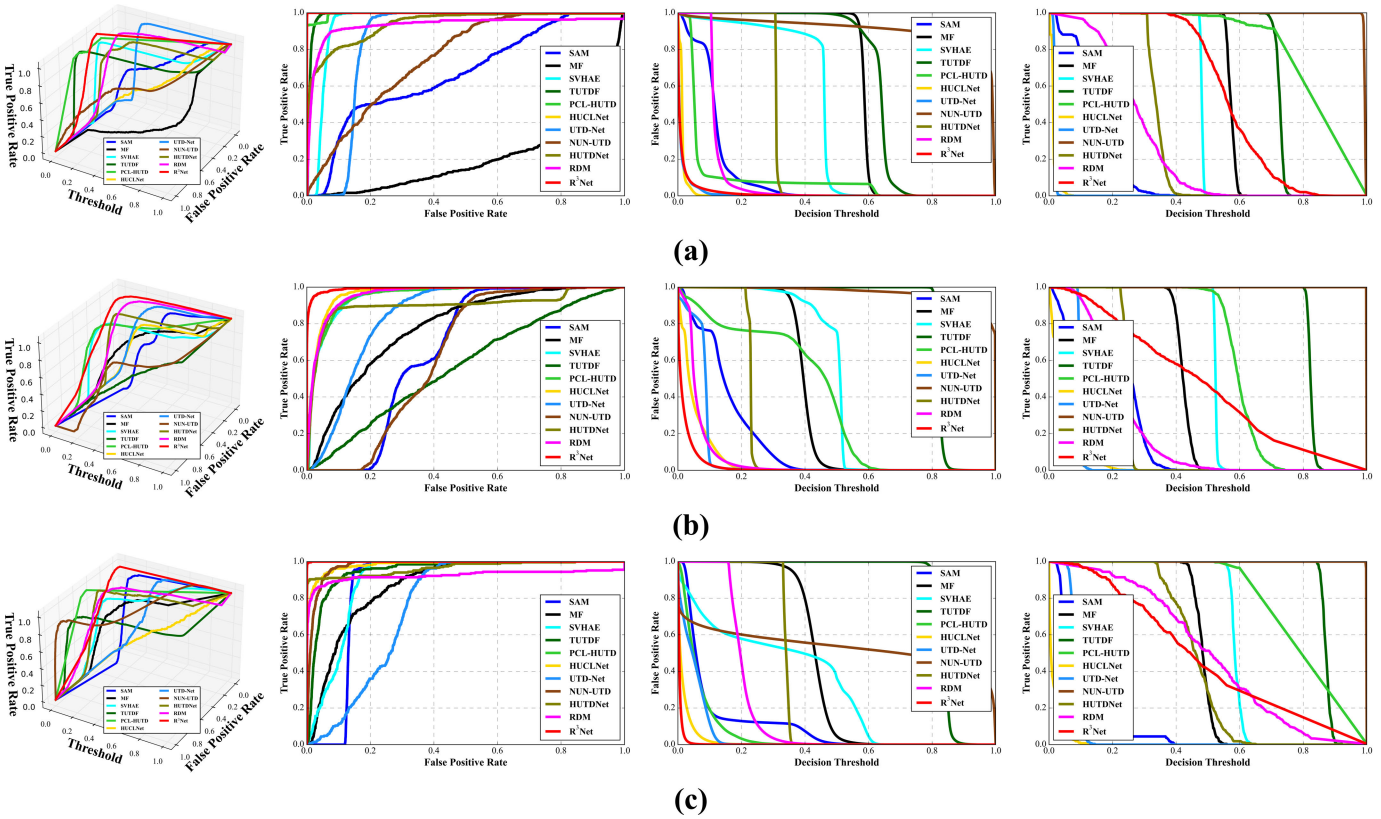


Fig. 5. ROC-based evaluation results. (a) Lake-Scene 1, (b) River-Scene 1, and (c) Sea-Scene 1. (From left to right) 3-D ROC surface, 2-D FPR–TPR curve, 2-D τ–TPR curve, and 2-D τ–FPR curve.

background suppression, producing diffuse response patterns with substantial background confusion and thus failing to reliably highlight the weak underwater targets.

Separation-based approaches show improved response regularity at the structural level. UTD-Net effectively suppresses widespread background responses, but its activations over some target regions appear relatively weak. NUN-UTD produces more spatially continuous responses over target areas while also exhibiting extensive activations in visually homogeneous water regions. HUTDNet achieves a more balanced suppression of background responses, although fine structural details of targets become less prominent in certain scenes.

In comparison, RDM concentrates responses on a limited number of regions that deviate from the dominant background patterns, leading to cleaner background suppression, while extremely weak target signals may be partially attenuated. In contrast, R^{3}Net generates high-contrast and spatially coherent detection maps with consistently low background responses. The detected target responses are spatially well aligned with the ground-truth annotations, and background interference is effectively suppressed across the lake, river, and sea scenarios.

Overall, the visual results indicate that R^{3}Net provides clearer target–background contrast and higher spatial consistency than the competing methods in complex underwater environments, including scenes characterized by turbidity, cluttered seabed textures, and weak target signatures.

2) ROC Curve: To evaluate detection performance under varying thresholds, we report both 3-D and 2-D ROC curves. Fig. 5 presents ROC evaluations on one representative scene from the ATR2-HUTD-Lake subset. Results for the remaining scenes in the *River* and *Sea* subsets, together with detailed analyses, are provided in Section II-B.

a) 3-D ROC surfaces: The 3-D ROC surfaces illustrate the joint variation of true positive rate (TPR) and false positive rate (FPR) with respect to the decision threshold τ. SAM and MF form low-elevation surfaces across all subsets, indicating limited true-positive responses over a wide range of thresholds. Prediction-driven methods (TUTDF, PCL-HUTD, and HUCLNet) exhibit higher TPR regions, although some of these regions are accompanied by noticeable expansions along the FPR dimension, particularly for TUTDF. Separation-based models present more compact surface structures: UTD-Net yields well-shaped surfaces with comparatively lower TPR elevation, while NUN-UTD shows steep increases in both TPR and FPR over narrow threshold intervals. RDM produces smoother surfaces with more gradual transitions, and R^{3}Net consistently exhibits the steepest and most concentrated high-elevation regions, indicating a favorable joint TPR–FPR behavior across thresholds.

b) 2-D FPR–TPR curves: The ROC curves highlight the tradeoff between recall and false-alarm rate. SAM and MF saturate early with relatively low TPR. Among prediction-driven methods, TUTDF shows improved recall accompanied by a faster increase in FPR, whereas PCL-HUTD maintains a more competitive TPR under low-FPR conditions. UTD-Net achieves lower FPR but with reduced recall, while NUN-UTD

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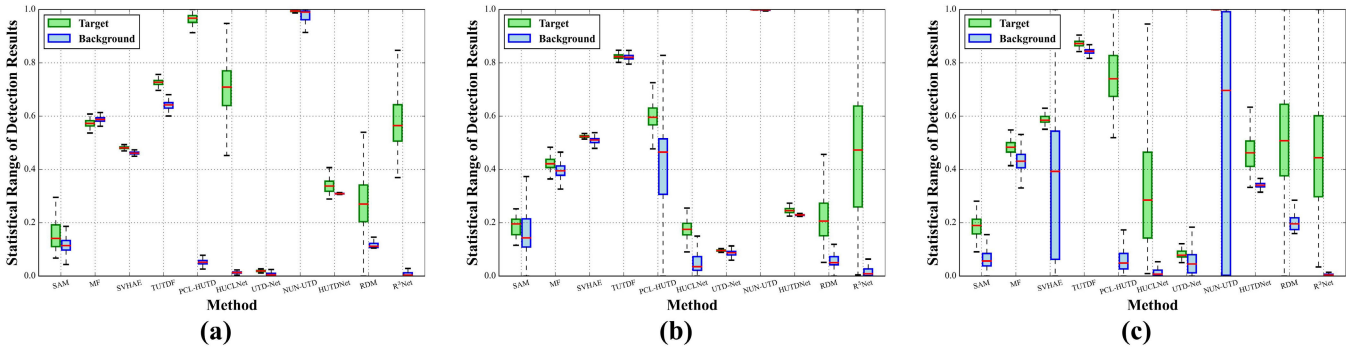


Fig. 6. Statistical distribution of detection scores for target pixels (green) and background pixels (blue) across six subscenes. (a) Lake-Scene1, (b) River-Scene1, and (c) Sea-Scene1. Greater separation between boxplots indicates higher discriminability between target and background regions.

exhibits concurrent increases in both TPR and FPR. HUTDNet provides a moderately controlled tradeoff. RDM demonstrates a balanced ROC profile, and R^{3}Net consistently maintains curves closest to the ideal top-left region across all subsets.

c) 2-D τ–FPR curves: These curves describe how background responses evolve with respect to the decision threshold. NUN-UTD shows a slow decay of FPR as τ increases, whereas UTD-Net suppresses FPR more effectively at lower thresholds. PCL-HUTD and HUCLNet exhibit segments with low FPR at relatively high thresholds, reflecting conservative operating regions. RDM gradually reduces FPR with increasing τ, while R^{3}Net sustains consistently low FPR over a wide threshold range, indicating stable background suppression.

d) 2-D τ–TPR curves: The τ–TPR curves characterize recall behavior under varying decision thresholds. TUTDF and NUN-UTD maintain high TPR across low-threshold regions, while their recall decreases more rapidly as τ increases. UTD-Net exhibits lower but more selective recall across thresholds. RDM achieves moderate yet stable TPR decay, whereas R^{3}Net maintains high recall in mid-range thresholds and exhibits smooth degradation as τ increases, reflecting stable detection performance under varying operating conditions.

e) Summary: Overall, the combined analyses of 3-D ROC surfaces and 2-D threshold-dependent curves indicate that while several methods achieve competitive performance in specific operating regions, their behaviors are often sensitive to threshold selection. In contrast, R^{3}Net demonstrates consistently favorable TPR–FPR tradeoffs, stable background suppression, and smooth recall variation across thresholds, indicating robust performance across diverse operating conditions.

3) TBS Map: TBS maps assess the contrast between target and background detection responses, providing a complementary perspective to ROC-based metrics. The score distributions are visualized using boxplots, where green and blue boxes denote target and background responses, respectively, and clearer separation reflects stronger distributional contrast. Fig. 6 presents representative results on the ATR2-HUTD subsets, while complete results are provided in Section II-C.

Traditional HTD methods exhibit substantial overlap between target and background score distributions, indicating limited separability. Prediction-driven HUTD methods (TUTDF and PCL-HUTD) increase target response levels but simultaneously elevate background scores, resulting in reduced target–background contrast and less selective responses across subsets. Separation-based models show more diverse behaviors: UTD-Net effectively suppresses background scores but also attenuates target responses, whereas NUN-UTD presents extremely wide background distributions with pronounced outliers, indicating high variability in background responses. RDM concentrates background scores within a narrow low-value range while maintaining moderate target activation, leading to a clearer distributional separation.

R^{3}Net achieves the most distinct target–background contrast, characterized by compact and consistently low background responses together with high and well-clustered target activations across all subsets. These observations indicate that R^{3}Net maintains stable pixel-level target–background separation under varying underwater environments.

Overall, the TBS analysis demonstrates that R^{3}Net consistently preserves the strongest statistical contrast between target and background responses across different underwater scenes.

4) AUC Value: We further perform quantitative evaluation using AUC-based metrics derived from ROC analysis, which provide scalar indicators of discriminability, recall stability, background suppression, and signal-to-noise robustness. In detail, we report $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$, together with two composite scores, AUC_{OA} and AUC_{SNPR}, offering a compact summary of overall performance in Table II.

a) $\text{AUC}_{(D,F)}$ value: This metric measures the separability between target and background responses. Traditional HTD methods consistently yield low scores, with MF dropping to 0.190 in *Lake-Scene2*, which indicates poor discrimination when target signatures are weak and easily mixed with surrounding water. Prediction-driven HUTD methods markedly increase $\text{AUC}_{(D,F)}$ by activating large spatial regions. For example, TUTDF and HUCLNet exceed 0.95 in most scenes, and PCL-HUTD reaches 0.997 and 0.995 in *Sea-Scene1* and *Sea-Scene2*, respectively. Separation-based approaches also achieve competitive values, such as NUN-UTD with 0.979 in *Sea-Scene1*, but their reliability is later constrained by unstable background control. In contrast, RDM and R^{3}Net maintain consistently high scores across all subsets, with R^{3}Net reaching 0.997 in *Lake-Scene2* and 0.994 in *Sea-Scene2*, confirming its robust target–background discrimination.

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TABLE II
QUANTITATIVE COMPARISON RESULTS. THE BEST AND SECOND-BEST RESULTS ARE IN BOLD AND WITH UNDERLINE

Method	ATR2-HUTD-Lake Scene1					ATR2-HUTD-Lake Scene2				
	$AUC_{(D,F)}\uparrow$	$AUC_{(F,\tau)}\downarrow$	$AUC_{(D,\tau)}\uparrow$	$AUC_{OA}\uparrow$	$AUC_{SNPR}\uparrow$	$AUC_{(D,F)}\uparrow$	$AUC_{(F,\tau)}\downarrow$	$AUC_{(D,\tau)}\uparrow$	$AUC_{OA}\uparrow$	$AUC_{SNPR}\uparrow$
SAM	0.670	0.119	0.119	0.670	0.994	0.480	0.069	0.079	0.491	1.155
MF	0.201	0.588	0.573	0.187	0.976	0.190	0.545	0.519	0.164	0.953
SVHAE	0.948	0.434	0.481	0.995	1.110	0.994	0.437	0.576	1.133	1.318
TUTDF	0.99	0.634	0.726	1.081	1.145	0.998	0.461	0.768	1.306	1.667
HUCLNet	0.995	0.209	0.71	1.501	3.393	**0.999**	0.265	0.765	1.500	2.891
PCL-HUTD	0.992	0.092	0.861	**1.761**	9.346	**0.999**	0.112	0.876	1.763	7.823
UTD-Net	0.846	**0.013**	0.019	0.853	1.492	0.944	0.041	0.073	0.976	1.771
NUN-UTD	0.758	0.913	**0.994**	0.838	1.088	0.765	0.792	**0.995**	0.968	1.256
HUTDNet	0.939	0.311	0.340	0.968	1.094	0.996	0.329	0.447	1.115	1.360
RDM	0.932	0.123	0.273	1.082	2.216	0.989	0.071	0.373	1.298	5.268
R^3Net	**0.996**	0.015	0.578	1.563	**39.31**	0.997	**0.006**	0.791	**1.785**	**135.6**

Method	ATR2-HUTD-River Scene1					ATR2-HUTD-River Scene2				
	$AUC_{(D,F)}\uparrow$	$AUC_{(F,\tau)}\downarrow$	$AUC_{(D,\tau)}\uparrow$	$AUC_{OA}\uparrow$	$AUC_{SNPR}\uparrow$	$AUC_{(D,F)}\uparrow$	$AUC_{(F,\tau)}\downarrow$	$AUC_{(D,\tau)}\uparrow$	$AUC_{OA}\uparrow$	$AUC_{SNPR}\uparrow$
SAM	0.657	0.156	0.210	0.711	1.349	0.656	0.061	0.074	0.669	1.210
MF	0.780	0.396	0.423	0.807	1.067	0.659	0.548	0.556	0.667	1.014
SVHAE	0.955	0.490	0.525	0.990	1.071	0.965	0.563	0.583	0.985	1.035
TUTDF	0.568	0.822	0.824	0.570	1.003	0.659	0.356	0.363	0.667	1.022
HUCLNet	0.966	0.055	0.175	1.086	3.206	0.924	0.178	0.327	1.073	1.837
PCL-HUTD	0.954	0.383	0.720	1.291	1.879	0.976	0.357	0.789	1.408	2.210
UTD-Net	0.843	0.080	0.096	0.860	1.208	0.889	0.075	0.088	0.903	1.176
NUN-UTD	0.632	0.968	**0.999**	0.663	1.032	0.503	0.977	**0.979**	0.505	1.002
HUTDNet	0.900	0.229	0.246	0.917	1.073	0.920	0.203	0.217	0.934	1.072
RDM	0.961	0.064	0.220	1.117	3.425	0.965	0.049	0.142	1.059	2.923
R^3Net	**0.984**	**0.021**	0.469	**1.432**	**22.33**	**0.987**	**0.011**	0.457	**1.443**	**41.64**

Method	ATR2-HUTD-Sea Scene1					ATR2-HUTD-Sea Scene2				
	$AUC_{(D,F)}\uparrow$	$AUC_{(F,\tau)}\downarrow$	$AUC_{(D,\tau)}\uparrow$	$AUC_{OA}\uparrow$	$AUC_{SNPR}\uparrow$	$AUC_{(D,F)}\uparrow$	$AUC_{(F,\tau)}\downarrow$	$AUC_{(D,\tau)}\uparrow$	$AUC_{OA}\uparrow$	$AUC_{SNPR}\uparrow$
SAM	0.866	0.101	0.077	0.842	0.757	0.819	0.057	0.055	0.817	0.961
MF	0.872	0.430	0.484	0.926	1.125	0.828	0.339	0.348	0.837	1.027
SVHAE	0.902	0.316	0.588	1.174	1.861	0.859	0.368	0.401	0.892	1.091
TUTDF	0.952	0.841	0.872	0.984	1.037	0.903	0.426	0.482	0.959	1.131
HUCLNet	0.985	0.019	0.325	1.292	17.50	0.989	0.053	0.474	1.420	8.857
PCL-HUTD	**0.997**	0.062	0.792	**1.728**	12.84	**0.995**	0.229	0.809	**1.576**	3.538
UTD-Net	0.762	0.050	0.083	0.796	1.678	0.774	0.043	0.070	0.801	1.631
NUN-UTD	0.979	0.534	**0.999**	1.444	1.871	0.975	0.959	**0.984**	0.999	1.025
HUTDNet	0.966	0.341	0.458	1.083	1.343	0.974	0.269	0.382	1.087	1.420
RDM	0.924	0.204	0.503	1.223	2.463	0.889	0.166	0.381	1.104	2.291
R^3Net	0.992	**0.006**	0.481	1.475	**74.42**	0.994	**0.014**	0.518	1.504	**36.55**

b) $AUC_{(F,\tau)}$ value: This metric evaluates the stability of background responses under varying thresholds, where lower values indicate stronger suppression. Traditional methods like MF produce large scores, e.g., 0.913 in *River-Scene1*, showing high sensitivity to noise. SVHAE and HUTDNet maintain moderate values, while TUTDF and PCL-HUTD

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TABLE III
QUANTITATIVE RESULTS OF ABLATION STUDIES ON THE ATR2-HUTD DATASET

Module Name	Design	$AUC_{(D,F)}\uparrow$	$AUC_{(F,\tau)}\downarrow$	$AUC_{(D,\tau)}\uparrow$	$AUC_{OA}\uparrow$	$AUC_{SNPR}\uparrow$	Training Time(h)↑
Full Model	N/A	0.992	0.012	0.549	1.534	58.31	0.325
RGSR Module	w/o Attn-Sparsity	0.926	0.311	0.499	1.114	1.605	0.358
	w/o Sparsity	0.891	0.431	0.453	0.913	1.051	0.391
	w/o Repulsion	0.797	0.105	0.184	0.876	1.752	0.317
	w/o RGSR	0.682	0.595	0.132	0.219	0.222	1.783
PARR Module	w/o MaskingOpt	0.973	0.088	0.512	1.397	5.818	0.343
	w/o PARR	0.943	0.113	0.315	1.147	3.098	0.308

produce larger $AUC_{(F,\tau)}$ scores, suggesting overactivation in background regions. UTD-Net achieves some of the lowest background accumulations, including 0.013 in *Lake-Scene1* and 0.041 in *Lake-Scene2*, whereas NUN-UTD generates the highest values across all subsets, with 0.913 in *Lake-Scene1* and 0.968 in *River-Scene1*, indicating severe instability. In particular, R^{3}Net reaches 0.006 in *Lake-Scene2* and 0.011 in *River-Scene2*, confirming stable false-positive suppression across confidence levels.

c) $AUC_{(D,\tau)}$ value: This metric measures the robustness of target recall under varying thresholds. SAM performs poorly, often below 0.1 in complex scenes (e.g., *Lake-Scene2*, *River-Scene2*). NUN-UTD achieves near-perfect scores, such as 0.995 in both *Lake-Scene2* and *River-Scene1*, but this is mainly due to overestimating background regions. R^{3}Net maintains balanced performance, reaching 0.791 in *Lake-Scene2* and 0.518 in *Sea-Scene2*, showing sensitivity to true targets while preserving stability.

d) AUC_{OA} value: This composite metric reflects the joint effects of separability, recall stability, and false-alarm suppression. RDM performs well, with 1.082 in *Lake-Scene1* and 1.117 in *River-Scene1*, validating residual-based detection. R^{3}Net achieves even higher scores, reaching 1.563 and 1.432 in the same scenes, demonstrating the benefit of physics-aware reasoning. In contrast, methods such as TUTDF and NUN-UTD show inconsistent results due to imbalance between recall and background suppression.

e) AUC_{SNPR} value: As a signal-to-noise indicator, this metric highlights the ability to sustain discriminability while minimizing false alarms. R^{3}Net achieves outstanding results, with 135.6 in *Lake-Scene2* and 41.64 in *River-Scene2*, far exceeding all baselines that often remain below 2. These values demonstrate strong resilience to weak signals and noisy backgrounds. RDM also performs well but shows less consistency across scenes.

f) Summary: Across the dataset, R^{3}Net outperforms all HTD and HUTD baselines on every metric. It achieves higher $AUC_{(D,F)}$ and $AUC_{(D,\tau)}$ while maintaining lower $AUC_{(F,\tau)}$, resulting in superior composite scores in both AUC_{OA} and AUC_{SNPR}. These results confirm its effectiveness and generalizability in diverse underwater environments.

C. Ablation Study

We conduct ablation experiments on ATR2-HUTD to evaluate the contribution of each component in R^{3}Net. The performance under each ablation setting is assessed by averaging results across all scenes using five AUC-based metrics, which together measure TBS, background suppression, and robustness under signal degradation. The results are summarized in Table III.

1) Effect of Rejection-Guided Spectral Reconstruction Module: The rejection-guided spectral reconstruction (RGSR) module reconstructs hyperspectral observations in a sparse and interpretable form. It emphasizes target-relevant signals and suppresses background confusion through tailored regularizations. The attention-based sparsity constraint focuses the model on informative regions. The general sparsity constraint prevents redundant abundance activation. The repulsion constraint promotes endmember separation to reduce spectral interference. To evaluate the contribution of each design, we test four ablation settings.

1) *w/o Attn-Sparsity:* The attention-guided sparsity term in (6) is removed, while the general ℓ_1 sparsity constraint is retained to enforce global compactness.
2) *w/o Sparsity:* All sparsity regularization terms are disabled, allowing the model to activate a large number of spectral components without restriction.
3) *w/o Repulsion:* The repulsion term is removed, which weakens the enforcement of distinctiveness between the learned spectral endmembers and the reference spectrum.
4) *w/o RGSR:* The entire rejection-guided reconstruction process is discarded, and the subsequent physics-aware residual reasoning is initialized directly with the random detection map.

The results in Table III confirm the necessity of each RGSR component. Removing the attention-based sparsity constraint (w/o Attn-Sparsity) reduces $AUC_{(D,F)}$ from 0.992 to 0.926, showing that ordinary ℓ_1 sparsity is insufficient for fine-grained background suppression. Disabling all sparsity constraints (w/o Sparsity) further decreases discriminability ($AUC_{(D,F)}$ = 0.891) and weakens background stability ($AUC_{(F,\tau)}$ = 0.431), confirming the key role of sparsity in reducing false posi-

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tives. Excluding the repulsion term (w/o Repulsion) drastically lowers $AUC_{(D,\tau)}$ from 0.549 to 0.184, indicating that weak targets cannot be preserved when endmembers lose distinctiveness from background spectra. Eliminating the entire RGSR module (w/o RGSR) causes the most severe degradation (e.g., $AUC_{(D,F)} = 0.682$, $AUC_{OA} = 0.219$, and $AUC_{SNPR} = 0.222$), since the subsequent residual reasoning process fails to converge without reliable initialization. These findings confirm that RGSR is indispensable, as it enhances robustness, ensures optimization stability, and improves spectral interpretability in underwater conditions with low TBS. Moreover, the training time increases moderately when removing RGSR, indicating that RGSR also contributes to efficient model convergence.

2) Effect of Physics-Aware Residual Reasoning Module: The physics-aware residual reasoning (PARR) module refines the initial results from RGSR using a physics-aware residual reasoning strategy. It applies a masking-based optimization scheme that directs reasoning toward regions consistent with the image formation model, thereby suppressing residual background activations and enhancing detection reliability. To assess its contribution, we evaluate two settings.

1) *w/o MaskingOpt:* The masking-guided optimization is removed, and residual reasoning is conducted directly on threshold-selected pixels in a pixel-wise manner.
2) *w/o PARR Module:* The entire PARR module is removed, and the final detection output is obtained directly from the RGSR module.

Based on Table III, removing the masking-based optimization (w/o MaskingOpt) leads to moderate degradation. $AUC_{(D,F)}$ decreases from 0.992 to 0.973, $AUC_{(D,\tau)}$ from 0.549 to 0.512, and $AUC_{(F,\tau)}$ rises from 0.012 to 0.088. Consequently, AUC_{OA} and AUC_{SNPR} decline from 1.534 to 1.397 and from 58.31 to 5.818, respectively, showing that pixel-wise optimization with threshold tuning is suboptimal since it ignores spatial context and physical priors. Further removing the entire PARR module (w/o PARR) results in more severe degradation. Although $AUC_{(D,F)}$ remains 0.943 due to RGSR guidance, $AUC_{(F,\tau)}$ drops to 0.113 and $AUC_{(D,\tau)}$ drops to 0.315, leading to an overall decline ($AUC_{OA} = 1.147$ and SNPR = 3.098). These results highlight the critical role of PARR in suppressing false positives and enhancing weak target visibility. In summary, the physics-aware residual reasoning module significantly improves discriminative power and robustness by enforcing physical consistency and enabling effective global optimization.

3) Main Conclusions From Ablation Studies: Based on the above ablation results, we further summarize the following key conclusions.

1) The RGSR module is indispensable to the whole framework, since removing it causes the most severe degradation in all metrics and even makes the subsequent residual reasoning process difficult to converge. This indicates that RGSR not only preserves weak target cues but also provides reliable initialization and optimization support for the following stage.
2) The PARR module is not an isolated postprocessing component, but a necessary physics-aware refinement of the learning-based detection network. Although RGSR alone can already preserve weak target candidates, removing PARR leads to clear degradation in false-alarm-related metrics, showing that the learning-based network is responsible for producing candidate weak-target responses, while PARR is responsible for verifying their physical plausibility and suppressing spurious activations.
3) The masking-based optimization within PARR is also necessary, since replacing it with pixel-wise threshold-based reasoning causes noticeable deterioration, especially in false-alarm suppression and robustness. This confirms that effective residual reasoning requires global optimization guided by physical priors, rather than simple local postprocessing.
4) Overall, these results demonstrate that RGSR and PARR are strongly complementary: the former enhances weak target preservation, while the latter improves physical reliability and false-alarm suppression. Their interaction is therefore essential to the effectiveness of the proposed framework.

D. Hyperparameter Sensitivity Analysis

To assess the robustness of R^3Net, we analyze three groups of hyperparameters from the RGSR and PARR modules: 1) the endmember number N in the RGSR module; 2) loss weights λ_s and λ_r in (10); and 3) regularization weights λ_c and α in (12) and (14).

1) Influence of Endmember Number N: We first examine the sensitivity of R^3Net to the number of endmembers N in the RGSR module. As shown in Fig. 7, performance varies across scenes. In the relatively homogeneous *Lake* environment, the best results occur at $N = 6$ ($AUC_{(D,F)} = 0.996$, $AUC_{OA} = 1.674$, and $AUC_{SNPR} = 87.46$). The more complex *River* scene benefits from larger bases, peaking at $N = 12$ ($AUC_{(D,F)} = 0.985$, $AUC_{OA} = 1.437$, and $AUC_{SNPR} = 31.98$). The *Sea* scene is optimal at $N = 8$ ($AUC_{(D,F)} = 0.993$, $AUC_{OA} = 1.489$, and $AUC_{SNPR} = 55.48$) and remains stable when N is between 6 and 10, indicating greater tolerance. Overall, AUC_{SNPR} is most sensitive to N, while the other metrics are robust around their optima. This shows that R^3Net is stable under moderate changes in N, but scene-specific adaptation yields benefits in low-SNR conditions.

2) Joint Influence of λ_s and λ_r: We next study the coupled effects of the sparsity weight λ_s and the repulsion weight λ_r. As shown in Fig. 8, R^3Net is robust across a broad range, with optimal performance at ($\lambda_s = 0.001, \lambda_r = 10$) ($AUC_{(D,F)} = 0.992$, $AUC_{OA} = 1.534$, and $AUC_{SNPR} = 58.31$). Nearby settings, such as (0.001, 7.5) or (0.0005, 10), cause only minor drops, while even extreme cases keep $AUC_{(D,F)}$ above 0.91 and AUC_{OA} above 1.39, though AUC_{SNPR} decreases to about 33. This shows that while exact tuning is not required, a reasonable balance between sparsity and repulsion is important for preserving weak target signals.

3) Joint Influence of λ_c and α: Finally, we evaluate the mask stability weight λ_c and abundance constraint weight α in the PARR module. The best configuration is ($\lambda_c = 0.005, \alpha = 5$), with $AUC_{(D,F)} = 0.992$, $AUC_{OA} = 1.534$,

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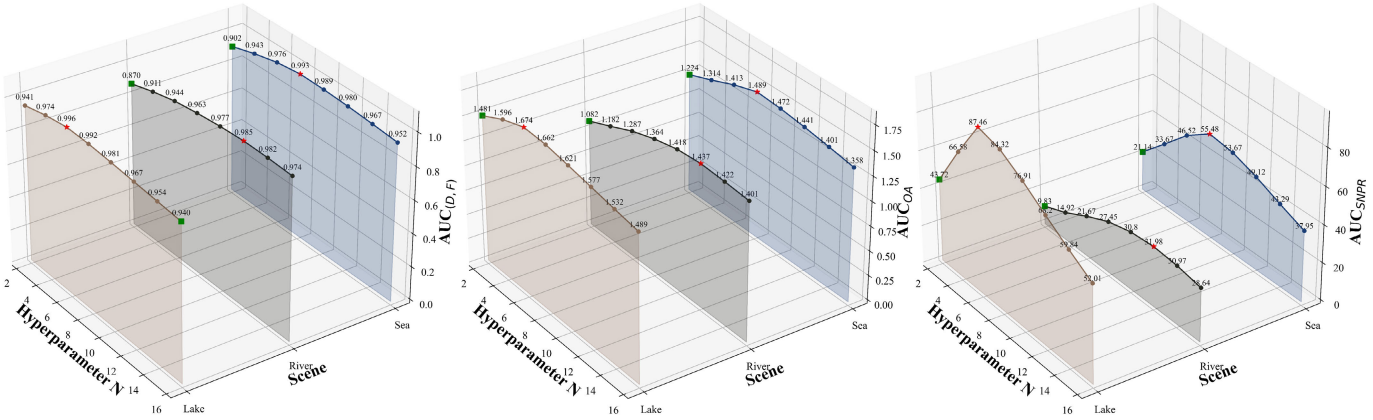


Fig. 7. Sensitivity analysis of R^{3}Net with respect to the number of endmembers N under three underwater scenes. The maximum and minimum AUC values are marked in red and green for each scene.

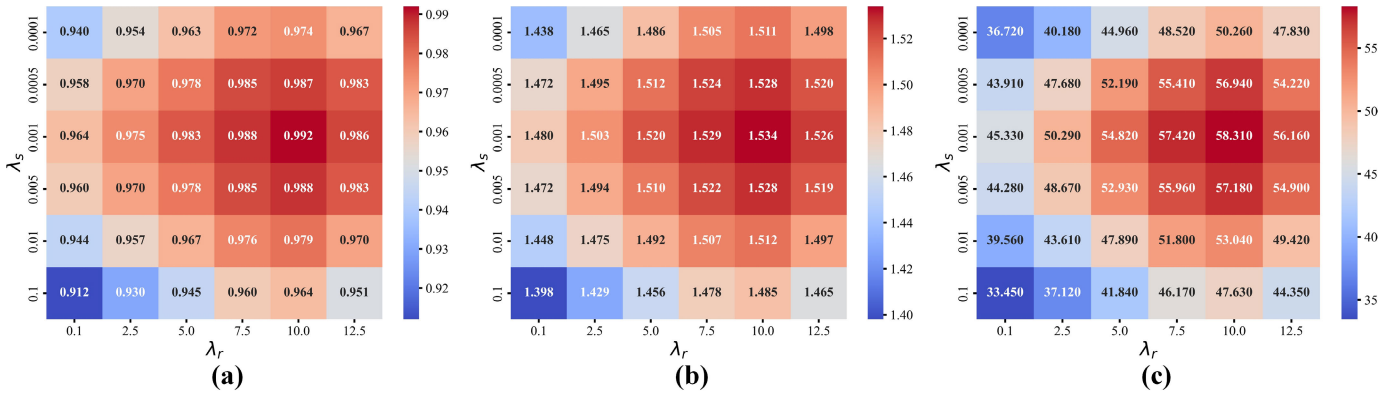


Fig. 8. Joint sensitivity analysis of sparsity coefficient λ_s and repulsion weight λ_r. (a) $\mathrm{AUC}_{(D,F)}$. (b) $\mathrm{AUC}_{\mathrm{OA}}$. (c) $\mathrm{AUC}_{\mathrm{SNPR}}$.

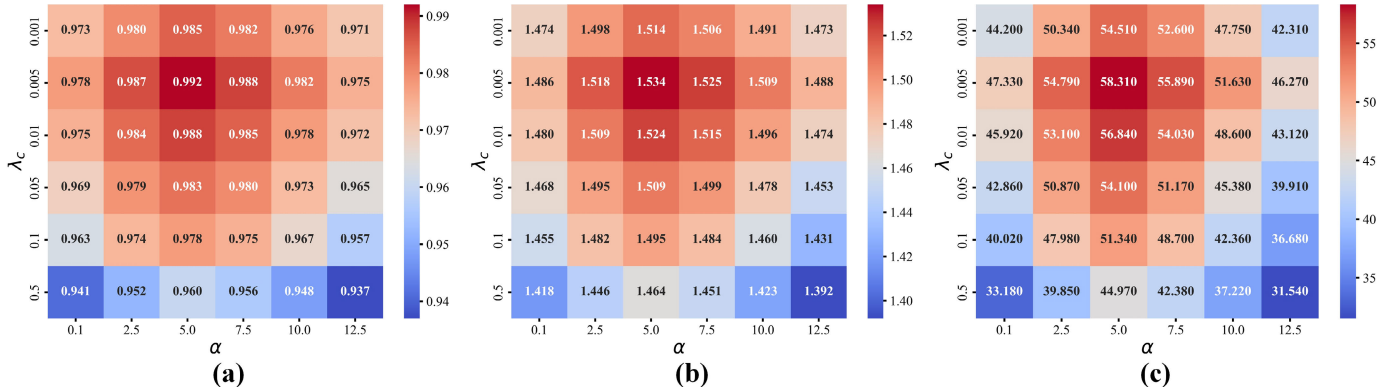


Fig. 9. Joint sensitivity analysis of confidence mask stability weight λ_c and abundance constraint strength α. (a) $\mathrm{AUC}_{(D,F)}$. (b) $\mathrm{AUC}_{\mathrm{OA}}$. (c) $\mathrm{AUC}_{\mathrm{SNPR}}$.

and $\mathrm{AUC}_{\mathrm{SNPR}} = 58.31$ (Fig. 9). Nearby settings, such as $(0.005, 2.5)$ or $(0.01, 5)$, give almost identical performance, with deviations within 0.005 in $\mathrm{AUC}_{(D,F)}$, confirming robustness to moderate changes. The model is more sensitive to α: very small values (e.g., 0.1) weaken spectral constraints and introduce noise, while very large ones (e.g., 12.5) restrict flexibility and reduce the ability to capture weak targets, both lowering $\mathrm{AUC}_{\mathrm{SNPR}}$ to around 33. In contrast, λ_c leads to smoother changes: larger values suppress false alarms but may also miss faint targets, yet overall detection remains stable. Overall, the broad high-performance plateau around $(0.005, 5)$ shows that R^{3}Net achieves reliable results without the need for precise tuning, although α requires more careful adjustment.

E. Efficiency Analysis

We analyze the computational efficiency of R^{3}Net compared to other methods in terms of training time, inference time, floating-point operations (FLOPs), and memory usage. The training and inference time are evaluated for runtime performance, while FLOPs and memory usage are assessed for computational complexity. All experiments are measured on

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TABLE IV
EFFICIENCY ANALYSIS OF THE COMPARED METHODS. THE BEST PERFORMANCES ARE HIGHLIGHTED IN BOLD

Method	Training Time (h)↓	Inference Time (s)↓	FLOPs (G)↓	Memory (MB)↓	AUC_{SNPR}↑
SAM	-	11.83	-	-	1.07
MF	-	**2.94**	-	-	1.03
SVHAE	5.94	20.67	5.91	0.95	1.25
TUTDF	0.75	8.37	742.4	3.18	1.17
HUCLNet	0.96	11.78	88.83	32.25	6.28
PCL-HUTD	0.77	3.01	64.42	0.55	6.27
UTD-Net	0.98	18.96	**3.75**	**0.076**	1.49
NUN-UTD	5.63	20.49	12.95	0.56	1.21
HUTDNet	0.46	6.98	43.86	3.28	1.23
RDM*	**0.31**	3.38	11.62	0.23	3.10
R^3Net*	0.33	4.17	12.37	0.23	**58.31**

* RDM and R^3Net have the similar computational efficiency, since PARR module in R^3Net only introduces negligible extra parameters.

seven NVIDIA A6000 GPUs using the ATR2-HUTD dataset. The results are summarized in Table IV, where the best performances are highlighted in bold.

From the perspective of computational complexity, R^3Net requires 12.37 G FLOPs, which is still relatively low compared with several representative deep learning-based competitors, such as TUTDF (742.4 G), HUCLNet (88.83 G), PCL-HUTD (64.42 G), and HUTDNet (43.86 G). Although UTD-Net achieves the lowest FLOPs among the compared learning-based methods (3.75 G), its detection performance is far inferior to that of R^3Net. This result indicates that the superior performance of R^3Net does not come from excessive computation, but from a more effective detection mechanism.

According to the results, we would like to emphasize that the proposed R^3Net achieves a favorable efficiency–performance tradeoff rather than relying on excessive computational cost. In terms of training efficiency, R^3Net requires only 0.33 h, which is very close to RDM (0.31 h) and is substantially lower than several representative learning-based methods such as NUN-UTD (5.63 h), UTD-Net (0.98 h), HUCLNet (0.96 h), and TUTDF (0.75 h). In terms of inference, R^3Net requires 4.17 s, which remains within a practical range and is only slightly higher than RDM (3.38 s), while achieving much better detection performance.

More importantly, the additional physics-aware residual reasoning module introduces only marginal overhead. Compared with RDM, R^3Net increases the training time by only 0.02 h and the inference time by only 0.79 s, while the memory cost remains the same at 0.23 MB, indicating that the PARR module adds negligible extra parameters. However, this very limited extra cost leads to a dramatic improvement in detection robustness, with AUC_{SNPR} increasing from 3.10 to 58.31.

Therefore, the proposed method does not sacrifice practicality for performance. Instead, it maintains low computational overhead while delivering substantially better detection capability than representative baseline methods. We believe these results demonstrate that R^3Net is not only effective but also efficient and practical for real-world weak underwater target detection scenarios.

VI. CONCLUSION

This article introduced R^3Net, a rejection-guided residual reasoning framework for HUTD. The method tackles two critical challenges: preserving weak target signals and reducing false alarms in complex aquatic environments. R^3Net integrates two key modules. The rejection-guided spectral reconstruction module adaptively models background end-members under sparsity and repulsion constraints, isolating residual spectra that capture target cues. The physics-aware residual reasoning module then refines these residuals through masking-based optimization guided by imaging priors, effectively suppressing false activations while enhancing low-contrast targets. Extensive experiments demonstrate that R^3Net consistently outperforms SOTA baselines, achieving high accuracy, robust background suppression, and strong resilience to signal degradation. Ablation and sensitivity analyses confirm the necessity of each module and the stability of the framework under varied hyperparameter settings. In summary, R^3Net establishes a separation-based paradigm that unifies residual reasoning, physical priors, and adaptive optimization, which provides a reliable solution for practical underwater remote sensing task.

Although the proposed framework achieves encouraging performance, several limitations still merit further investigation. First, the current physics-aware reasoning mechanism is built upon a simplified underwater imaging formulation, which may still be insufficient to fully capture more complex environmental variations in highly dynamic underwater scenes. Second, the efficiency of the proposed method may be further improved, as the current optimization process involves multiple iterations and may not be suitable for real-time applications in resource-constrained underwater platforms. Third, the present validation is mainly conducted on nearshore hyperspectral weak target detection scenarios, and its generalization to other weak-target detection tasks has yet to be further examined. These limitations also motivate future efforts toward improving robustness in more complex underwater environments, extending the framework to broader weak-target detection scenarios, and exploring its integration with more general foundation model-based representations.

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In light of the above limitations, several directions remain worthy of further investigation. Future work will focus on enhancing robustness under more complex and dynamically varying underwater environments, extending the proposed framework to other weak-target detection tasks with similar characteristics, and integrating it with more general foundation-model-based representations to further improve its generalization capability and practical applicability.

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